



Become our next #FutureStarter

Are you ready to make an impact? ZF is looking for talented individuals to join our team. As a #FutureStarter, you'll have the opportunity to shape the future of mobility. Join us and be part of something extraordinary!

About ZF

ZF is a global technology company that supplies systems for passenger cars, commercial vehicles, and industrial technology, driving the next generation of mobility. Our innovative solutions enable vehicles to see, think, and act across four key technology domains: Vehicle Motion Control, Integrated Safety, Automated Driving, and Electric Mobility.

We offer comprehensive product and software solutions for both established vehicle manufacturers and emerging transport and mobility service providers. ZF electrifies a wide range of vehicle types, contributing to reduced emissions, climate protection, and enhanced safety in mobility.

Job Description:

Research Intern – Knowledge Graphs and AI in Mechanical Engineering

Location: Hyderabad (On-site/Hybrid)

Duration: 1 Year

About the Role

We are seeking a highly motivated **Research Intern from IITs or IIITH or IISc** to join our AI Research team and work at the intersection of **Knowledge Graphs, Large Language Models (LLMs), and Mechanical Engineering**.

The objective of this internship is to develop intelligent systems that leverage **Knowledge Graphs and Generative AI** to enhance engineering workflows across the product lifecycle, including

mechanical design, simulation, manufacturing, production, quality, and engineering knowledge management.

The intern will research and build AI solutions that can capture engineering knowledge, reason over complex product data, support engineers in decision-making, and enable natural language interactions with engineering systems and repositories.

Key Responsibilities

Research

- Conduct research on the application of:
 - Knowledge Graphs (KGs), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Graph-Based Reasoning, Engineering AI Systems
- Study and reproduce state-of-the-art research relevant to:
 - Engineering Knowledge Representation ,Product Lifecycle Management (PLM), Digital Engineering, Design Automation, Manufacturing Intelligence, Industrial AI
- Investigate methods for combining Knowledge Graphs and LLMs for engineering problem solving.

Knowledge Graph Development

- Design and develop knowledge graphs representing:
 - Mechanical components, Product structures (BOMs), Design rules, Manufacturing processes, Simulation models, Engineering standards, Failure and quality data
- Develop ontology models for engineering domains.
- Build pipelines for extracting structured knowledge from:
 - CAD metadata, Engineering documents, Technical standards, Simulation reports, Manufacturing records
- Integrate graph databases and graph reasoning frameworks.

LLM Engineering

- Develop LLM-powered applications for engineering use cases.
- Build Graph-RAG systems that combine Knowledge Graphs with LLMs.
- Fine-tune domain-specific models using engineering datasets.
- Develop intelligent engineering assistants capable of:

- Design knowledge retrieval, Root-cause analysis, Engineering Q&A, Manufacturing support, Simulation insights
 - Evaluate and improve model performance for engineering tasks.
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Target Application Areas

The intern will contribute to AI solutions for:

Mechanical Design

- Design knowledge retrieval, Design rule validation, Component recommendation, BOM intelligence, Engineering change analysis, Requirements traceability

Simulation & Analysis

- Simulation data knowledge management, Correlation of simulation and test results, Failure mode analysis, Design optimization support, Engineering decision support systems

Manufacturing & Production

- Production troubleshooting, Process knowledge management, Manufacturing workflow intelligence, Quality analytics, Defect root-cause identification, Maintenance and operations knowledge systems
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Required Qualifications

- Currently pursuing **B.Tech, M.Tech, MS (Research), or PhD** at **IITs or IIITH or IISc** in:
 - Mechanical Engineering
 - Computational Engineering
 - Strong programming skills in Python.
 - Good understanding of:
 - Machine Learning, Artificial Engineering, Data Structures and Algorithms, Databases, Software Engineering Fundamentals
 - Familiarity with:
 - NLP and LLMs, Knowledge Representation, Graph Data Structures
 - Strong problem-solving and analytical skills.
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Preferred Qualifications

Candidates with experience in one or more of the following areas are highly encouraged to apply:

AI & LLMs

- Hugging Face Transformers, PyTorch, RAG Systems, Fine-Tuning Techniques, Agentic AI, Prompt Engineering

Knowledge Graphs

- Neo4j, RDF / OWL, Graph Databases, GraphRAG, Ontology Design, Semantic Search

Mechanical Engineering

- CAD Systems, PLM Platforms, Product Design, Simulation & CAE, Manufacturing Processes, Production Engineering

Data & Infrastructure

- SQL, Vector Databases, Azure AI Services, Cloud Platforms, Data Engineering Pipelines
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Research Areas

The intern will have opportunities to work on:

- Engineering Knowledge Graphs, Graph-RAG Architectures, LLMs for Mechanical Engineering, Design Intelligence Systems, Digital Engineering and PLM AI, Engineering Copilots, Knowledge-Based Manufacturing Systems, Simulation Knowledge Management, Industrial AI and Industry 4.0, Explainable AI for Engineering Applications
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Expected Outcomes

By the end of the internship, the candidate is expected to:

- Develop one or more engineering knowledge graphs for mechanical domains.
- Build Graph-RAG solutions for design, simulation, or production use cases.
- Create ontology frameworks for engineering knowledge representation.
- Develop AI assistants capable of answering complex engineering questions.
- Establish reusable pipelines for extracting and organizing engineering knowledge.
- Deliver research reports, prototype systems, technical publications, patents, or internal innovation assets.